

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Fourth Semester B. Tech (CBCS) Degree Examinations

Summer Semester July 2024

ME1103-1 – THEORY OF MACHINES
(ME)

Duration: 3 Hours

Max. Marks: 100

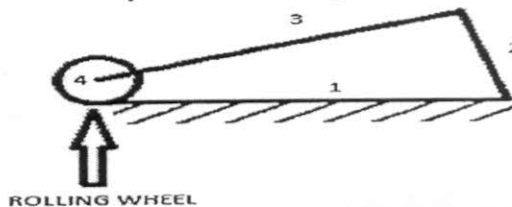
Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Find the mobility of the following mechanism with a higher pair.



- | | |
|---|---|
| A) 3 | B) -1 |
| C) 1 | D) 2 |
| 2. In a structure, the degree of freedom is | |
| A) 1 | B) 2 |
| C) Infinite | D) 0 |
| 3. In a mechanism, which one is true | |
| A) All links are fixed | B) One link is fixed |
| C) All the links are free | D) Two links are fixed |
| 4. A rigid body possesses _____ degrees of freedom. | |
| A) 1 | B) 2 |
| C) 4 | D) 6 |
| 5. The links of a Machine transmit | |
| A) Forces only | B) Motion only |
| C) Power and motion | D) Power only |
| 6. In a kinematic chain, a quaternary joint is equivalent to _____. | |
| A) one binary joint | B) Two binary joint |
| C) three binary joint | D) four binary joint |
| 7. In dynamic balancing following condition will hold | |
| A) Only the force polygon will be closed | B) Only the couple polygon will be closed |
| C) Both the force and couple polygon will be closed | D) Force polygon is closed but couple polygon is open |
| 8. For a static balancing | |
| A) The net dynamic force acting on the shaft is zero | B) The net couple acting on the shaft due to dynamic forces is zero |
| C) Net dynamic force and couple on the shaft is zero | D) Centre of gravity does not lie on the shaft axis |
| 9. A cam mechanism imparts | |
| A) Only Rotating motion | B) Only Oscillating motion |
| C) Only Reciprocating motion | D) Rotating motion, Oscillating motion and reciprocating motion. |
| 10. A cam and follower constitute a | |
| A) Higher pair | B) Lower Pair |
| C) Helical Pair | D) Spherical Pair |
| 11. In high speed engines the cam follower moves with | |
| A) S.H.M | B) Uniform velocity |
| C) Cycloidal motion | D) Uniform acceleration and retardation |

12. Cam follower generally used in automobile engines is
 A) Roller follower B) Knife edge follower
 C) Spherical faced follower D) Flat faced follower
13. A motor car moving at a certain speed takes a left turn in a curved path. If the engine rotates in the same direction as that of wheels, then due to the centrifugal forces
 A) the reaction on the inner wheels increases and on the outer wheels decreases B) the reaction on the outer wheels increases and on the inner wheels decreases
 C) the reaction on the front wheels increases and on the rear wheels decreases D) the reaction on the rear wheels increases and on the front wheels decreases
14. The axis of precession is _____ to the plane in which the axis of spin is going to rotate. If the gyroscopic effect is to be felt.
 A) Parallel B) Perpendicular
 C) Spiral D) Inclined
15. A disc is a spinning with an angular velocity of ω rad/s about the axis of spin. The couple applied to the disc causing precession will be
 A) $\frac{1}{2} I \omega^2$ B) $I \omega^2$
 C) $\frac{1}{2} I \omega \omega_p$ D) $I \omega \omega_p$
16. The planes of spin, precession and applied gyroscopic couple are
 A) The same one plane B) Two planes perpendicular to one another
 C) Three planes perpendicular to one another D) Three planes parallel to one another
17. The type of gears used to connect two parallel and coplanar shafts are
 A) Spur gears B) Helical gears
 C) Bevel gears D) Spiral gears
18. The module is the reciprocal of
 A) Circular pitch B) Diametral Pitch
 C) Pitch circle diameter D) Base circle diameter
19. Fluid resistance causes damping which is known as _____
 A) Resistance damping B) Fluid damping
 C) Viscous damping D) Liquid damping
20. The most salient features of any vibration are
 A) Amplitude & acceleration B) Frequency & acceleration
 C) Amplitude & frequency D) Velocity and acceleration

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

Marks BT* CO* PO*

1. a) Define the following terms
 I. Kinematic pair
 II. Kinematic chain
 III. Mechanism
 IV. Inversion
- b) Explain the Grashoff's Law for four bar mechanism.
- c) In a four link mechanism, torque T_4 has a magnitude 20Nm (CCW) respectively. Link lengths are AD=800mm, AB=300mm, BC=700mm, CD=400mm respectively. Find the torque to be applied on link 2 for the static equilibrium of the mechanism. Shown in figure1(c).

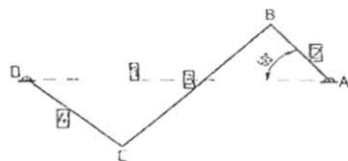


Figure 1(c).

2. a) Sketch and explain the inversions of Four bar Kinematic chain.

- b) For the static equilibrium of the mechanism shown in figure 2(b), determine the torque to be applied on link AB. Link lengths are as follows AB=20cm, BD=40cm.

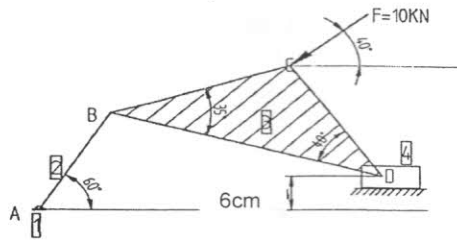


Figure 2(b).

8 L3 2 2

3. a) Define degrees of freedom. Determine the degrees of freedom for the mechanism shown in figure 3(a).



Figure 3(a).

8 L2 1 1

- b) A circular disc mounted on a shaft carries three attached masses of 4kg, 3kg and 2.5kg at radial distances of 75mm, 85mm and 50mm at angular positions of 45°, 135° and 240° respectively. The angular positions are measured counterclockwise from the reference line along the x-axis. Determine the amount of the counter mass at a radial distance of 75mm required for the static balance.

8 L3 2 2

Unit – II

4. a) A cam rotating clockwise at a uniform speed of 1000 r.p.m. is required to give a roller follower the motion defined below:

- Follower to move outwards through 50 mm during 120° of cam rotation,
- Follower to dwell for next 60° of cam rotation,
- Follower to return to its starting position during next 90° of cam rotation,
- Follower to dwell for the rest of the cam rotation.

The minimum radius of the cam is 50 mm and the diameter of roller is 10 mm. The line of stroke of the follower is off-set by 20 mm from the axis of the cam shaft. If the displacement of the follower takes place with uniform and equal acceleration and retardation on both the outward and return strokes, draw profile of the cam.

8 L3 3 2

- b) Derive an expression for Angle of Heel in case of a two-wheel automobile.

8 L2 3 1

5. a) A cam is to be designed for a knife edge follower with the following data:

- Cam lift = 40 mm during 90° of cam rotation with simple harmonic motion.
- Dwell for the next 30°.
- During the next 60° of cam rotation, the follower returns to its original position with simple harmonic motion.
- Dwell during the remaining 180°.

Draw the profile of the cam when the line of stroke is offset 20 mm from the axis of the cam shaft. The radius of the base circle of the cam is 40 mm.

8 L3 3 2

- b) Derive an expression for the gyroscopic couple for a plane disc spinning about its axis and having precessional motion about an axis perpendicular to the axis of spin.

8 L2 3 1

6. a) With a neat sketch explain the following terms w.r.t. cams

- Base circle
- Prime circle
- Pitch curve
- Pressure angle

8 L1 3 1

- b) An aero plane makes a complete quarter circle of 40m radius towards left when flying at 175km/hr. The mass of the rotary engine and propeller is 400kg with radius of gyration of 300mm. The engine rotates 2500rpm clockwise when viewed from the rear. Find the gyroscopic couple on the aircraft. What will be the effect if the aeroplane turn towards right instead of left?
- c) Find the angle of inclination with respect to the vertical of a two-wheeler negotiating a turn. Given: combined mass of the vehicle with its rider 250 kg; moment of inertia of the engine flywheel 0.3 kg-m^2 ; moment of inertia of each road wheel 1 kg-m^2 ; speed of engine flywheel 5 times that of road wheels and in the same direction; height of center of gravity of rider with vehicle 0.6m; two-wheeler speed 90 km/h; wheel radius 300 mm; radius of turn 50 m.

4 L3 3 2

4 L3 3 2

Unit – III

7. a) Define the following w.r.t involute tooth profile
i. Addendum ii) Dedendum iii) Circular pitch iv) Path of contact
iv) Contact ratio f) Module g) Diametral Pitch h) Total Depth
- b) Determine the natural frequency of the system shown in figure 7(a).

8 L1 4 1

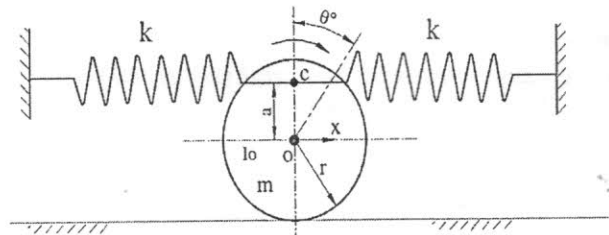


Figure 7(a)

8 L3 5 2

8. a) An epicyclic gear consists of three gears A, B and C as shown in figure 8(a) below. The gear A has 72 internal teeth and gear C has 32 external teeth. The gear B meshes with both A and C and is carried on an arm EF which rotates about the center of A at 18 r.p.m. If the gear A is fixed, determine the speed of gears B and C.

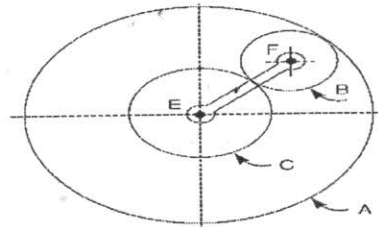


Figure 8(a)

8 L3 4 2

- b) Determine the natural frequency of the system shown in figure 8(b) below

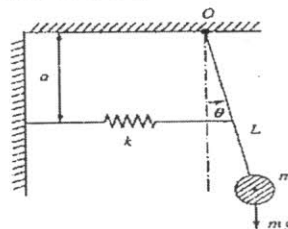


Figure 8(b)

8 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
